

Option D – Energy and the Environment

10. (a) The proton-proton chain accounts for the majority of the energy output from the Sun. The initial stage of the proton-proton chain is shown below.



- (i) State which force is responsible for this reaction. [1]

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- (ii) Determine the energy released by one such reaction. [2]

Masses: ${}_1^1\text{H} = 1.00728\text{u}$ ${}_1^2\text{H} = 2.01355\text{u}$
 $\text{e}^+ = 0.00055\text{u}$ $\nu_{\text{e}} = 0.000\ 00\text{u}$

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- (b) (i) Define the 'solar constant'. [2]

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- (ii) Suggest why the solar constant is not considered to be a constant by the scientific community. [1]

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- (iii) The temperature of the Sun's surface is 5780 K. Given that the surface area of the Sun is $6.1 \times 10^{18} \text{ m}^2$, determine the intensity of the Sun's radiation at a distance of $1.50 \times 10^{11} \text{ m}$ from its centre **and** state the name of a law used in your calculation. [4]

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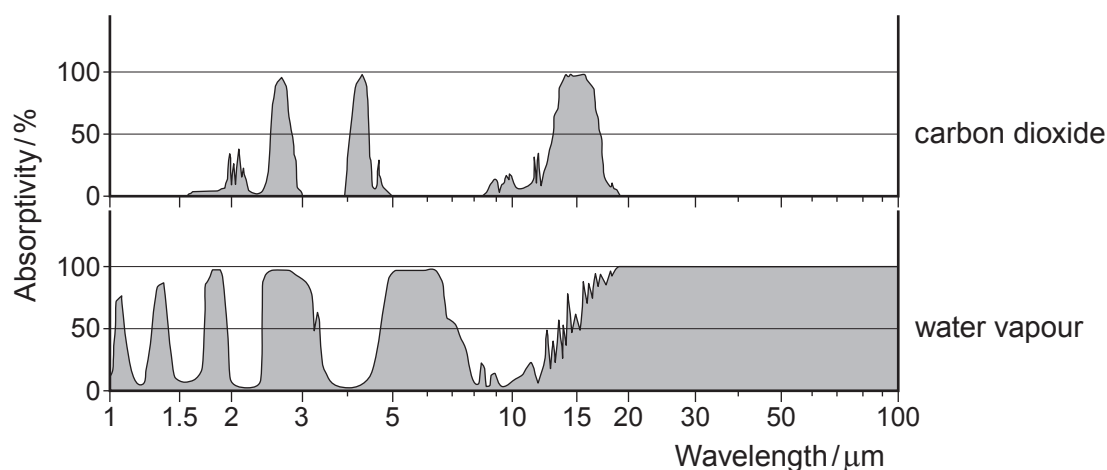
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- (c) The graph below shows the absorption spectra for carbon dioxide and water vapour as a function of the wavelength of the radiation incident on the gas in our atmosphere.



Studies show that the greenhouse effect is being enhanced by increased levels of carbon dioxide and water vapour. Use the graph to explain this, assuming the Earth emits radiation as a black body with a mean temperature of 288 K. [4]

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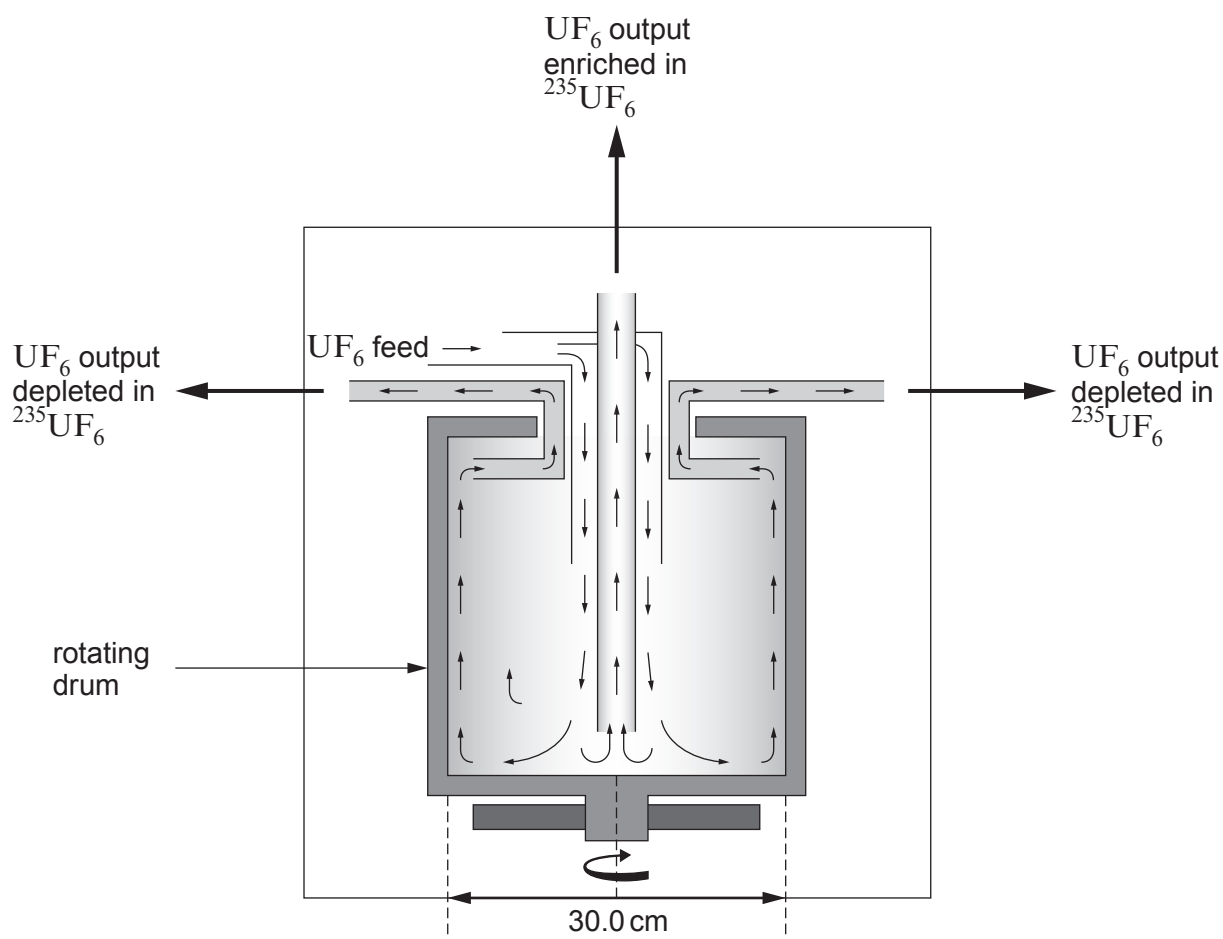
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- (d) Nuclear fission of uranium fuel is used on Earth in order to generate electrical energy. Natural uranium is composed of U-235 and U-238 nuclei.

- (i) Explain why enrichment of uranium is required before it is used in a fission reactor. [2]

- (ii) U-235 and U-238 are both reacted with fluorine to create $^{235}\text{UF}_6$ and $^{238}\text{UF}_6$ gas molecules. These gas molecules are then fed into a gas centrifuge consisting of a rotating drum where they undergo circular motion.



The diameter of the rotating drum is 30.0 cm and the speed of the outer surface of the rotating drum is 350 ms^{-1} . Calculate the number of revolutions per second made by the drum. [2]

- (iii) The output from the centrifuge has 1.15 times more $^{235}\text{UF}_6$ than was initially fed in. This enriched mixture is transferred into a series of identical centrifuges. Determine the number of centrifuges required to enrich the $^{235}\text{UF}_6$ in the gas mixture from 0.7 % to 5.0 %. [2]

END OF PAPER

